Harry Brook Runs Prediction

AI Course Project Proposal | FUUAST, Karachi

1. Selected Player

Harry Brook (English cricketer, right-handed batter)

2. Objective

Predict how many runs Harry Brook will score in his next One Day International (ODI) innings.

3. Approach

Supervised learning (the output is a numerical value: runs scored). This is a regression problem.

4. Model

Decision Tree Regressor (main model), compared with Random Forest Regressor.

5. Reason

- Supervised learning: each previous ODI innings has a known runs-scored value, so the data is labelled.
- Decision Tree: can use factors such as opponent, venue, batting position and recent form; it can model non-linear patterns and is relatively easy to interpret.
- Random Forest: combines multiple decision trees, which can reduce overfitting and provide more stable predictions.

6. Dataset (Authentic Sources)

Main source (ICC): Harry Brook player profile and match-by-match records
https://www.icc-cricket.com/rankings/66374/harry-brook

Verification source (ESPNcricinfo Statsguru): International cricket batting statistics and match records
https://stats.cricinfo.com/cricinfo/engine/stats/index.html

Additional official source (ECB): Player profile and playing role
https://www.ecb.co.uk/players/3375

ODI innings records will be collected with features such as opponent, venue, batting position, innings number and recent performance. The target column will be runs scored in each innings. Records will be checked against the ICC and ESPNcricinfo sources before model training.